

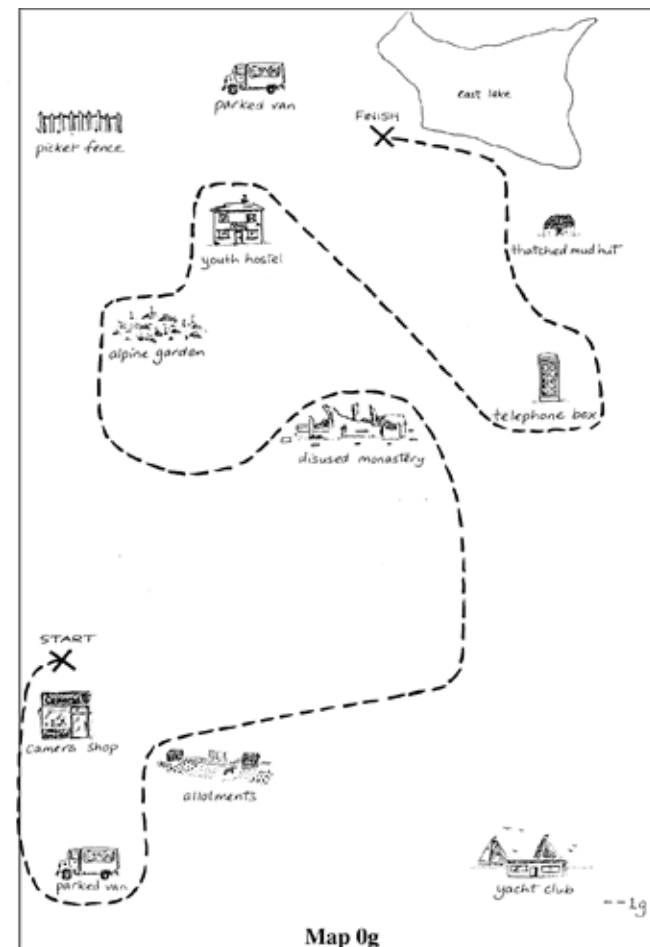
Seeing Is Not Sharing

Some Vision-Language Models Overestimate
Common Ground in Asymmetric Dialogue

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What could be shared $\neq$ what has been shared



Shared perception is not shared interpretation

POTENTIAL COMMON GROUND

Both participants could have access to the same kind of landmark.

Co-presence in the scene.

≠

ESTABLISHED COMMON GROUND

Built incrementally: the speaker refers, the addressee recognises or accommodates, both confirm or repair.

Clark & Wilkes-Gibbs 1986; Clark & Brennan 1991

A MODEL IS AN OVERHEARER

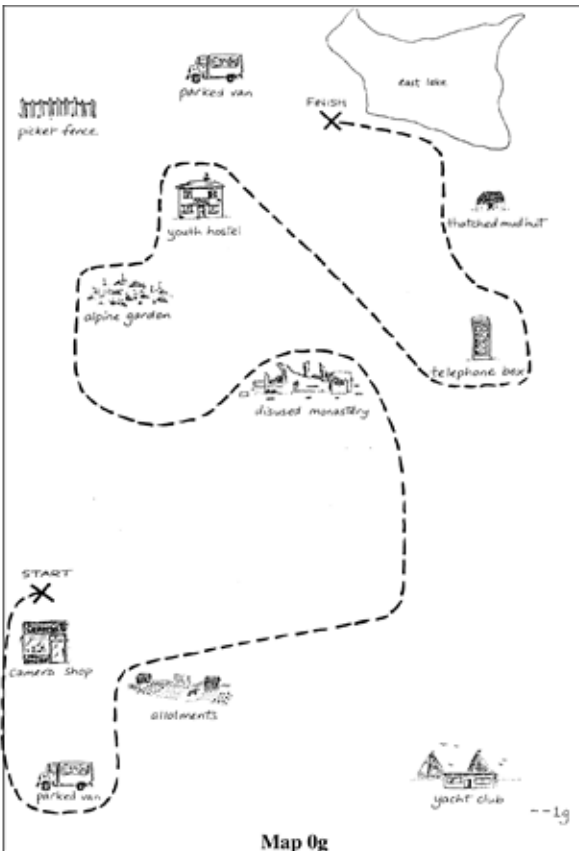
It sees the words, and sometimes the maps. It cannot ask a clarification question and it cannot repair.

Overhearers overestimate their own understanding (Schober & Clark 1989).

Can a vision-language model tell what could be shared from what has actually become shared?

This is the question the paper operationalises — and asymmetric dialogue is where the two come apart most cleanly.

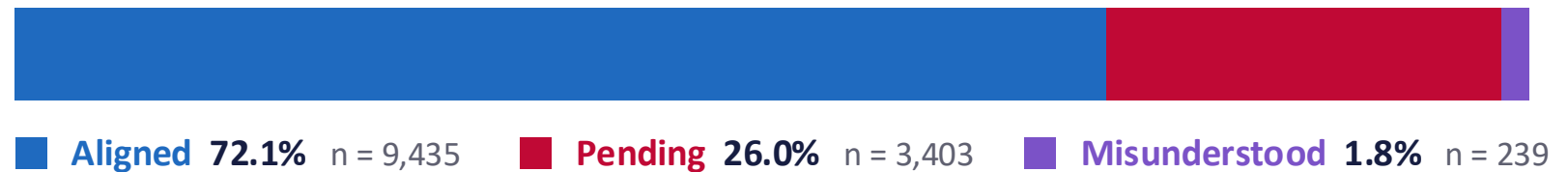
MapTask: two private maps that differ by design



- **Lexical variant** — the same landmark is named differently — “white water” vs. “rapids”
- **Existence** — a landmark is on one participant's map only
- **Multiplicity** — two parked vans on one map, one on the other

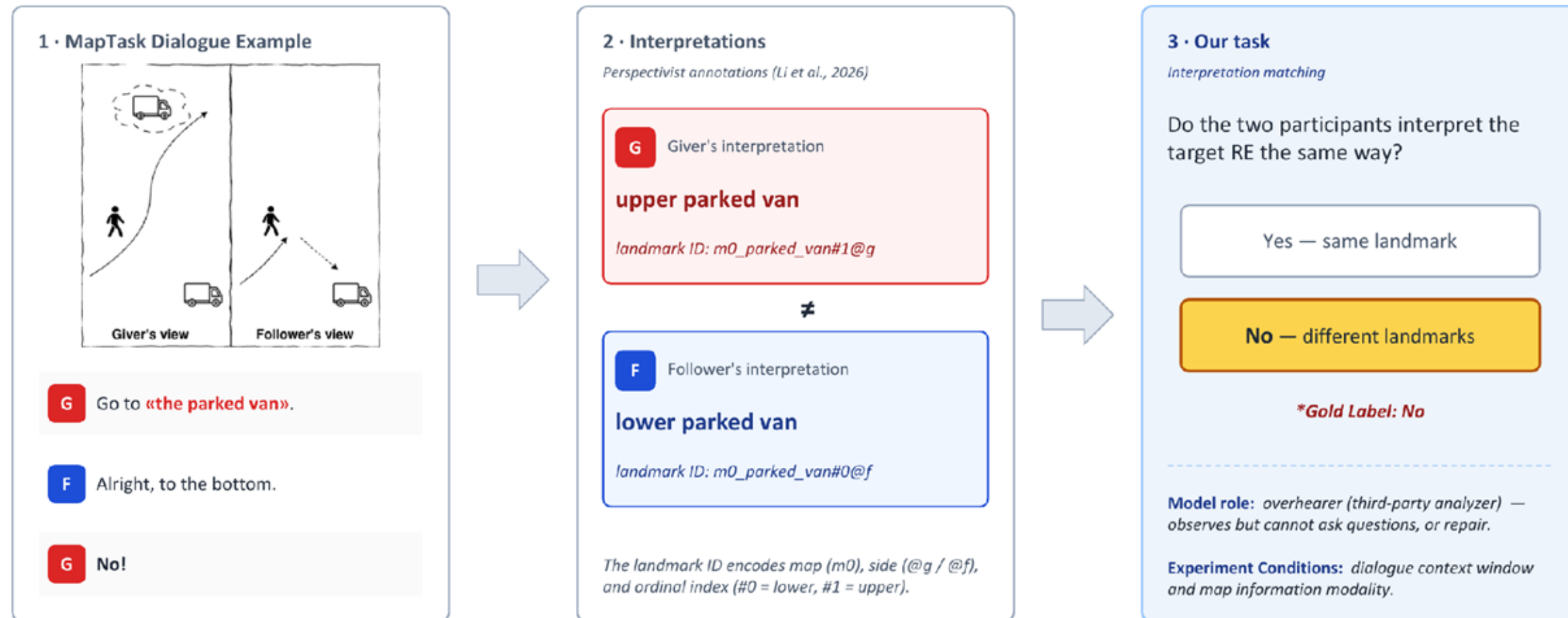
Perspectivist annotation (Li et al., LREC 2026): for every reference expression we know the giver's landmark and the follower's landmark, separately.

GOLD UNDERSTANDING STATE



THE TASK

Interpretation matching: do both participants mean the same landmark?



The model is an overhearer: it observes the evidence but cannot ask or repair. Gold label here is NO — and NO covers two cases: grounding is still pending, or it happened but to different landmarks.

Two independent variables, fully crossed, zero-shot

1 • DIALOGUE CONTEXT

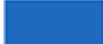
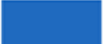
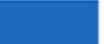
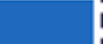
Four windows, from the current transaction up to the target line (curL) to the whole dialogue through the end of that transaction (startT).

Later turns carry confirmation and repair.

2 • MAP INFORMATION

Authentic maps (both / giver / follower); the same content delivered as text (landmark names, discrepancy descriptions); two content-free visual controls (blank grey, shuffled maps).

The controls are what separate content from channel.

	DIALOGUE CONTEXT WINDOW (small → large)			
MAP INFORMATION	curL	curT	startL	startT
No map (text-only)				
Both authentic maps				
Giver's map only				
Follower's map only				
Landmark names (text)				
Discrepancy description (text)				
Blank grey images				
Shuffled maps				
				<i>reported</i>

Five open VLMs: Qwen3-VL 2B / 4B / 8B and Gemma-3 4B / 12B. Qwen3-VL-8B runs the full grid; the other four run the four baseline map conditions, also across all four windows.

Decoding is constrained to a single YES / NO token — so the token probability is directly the model's confidence.

FINDING 1

Maps help — by pushing the model toward YES

MACRO-F1

.591 → **.671**

RECALL · ALIGNED

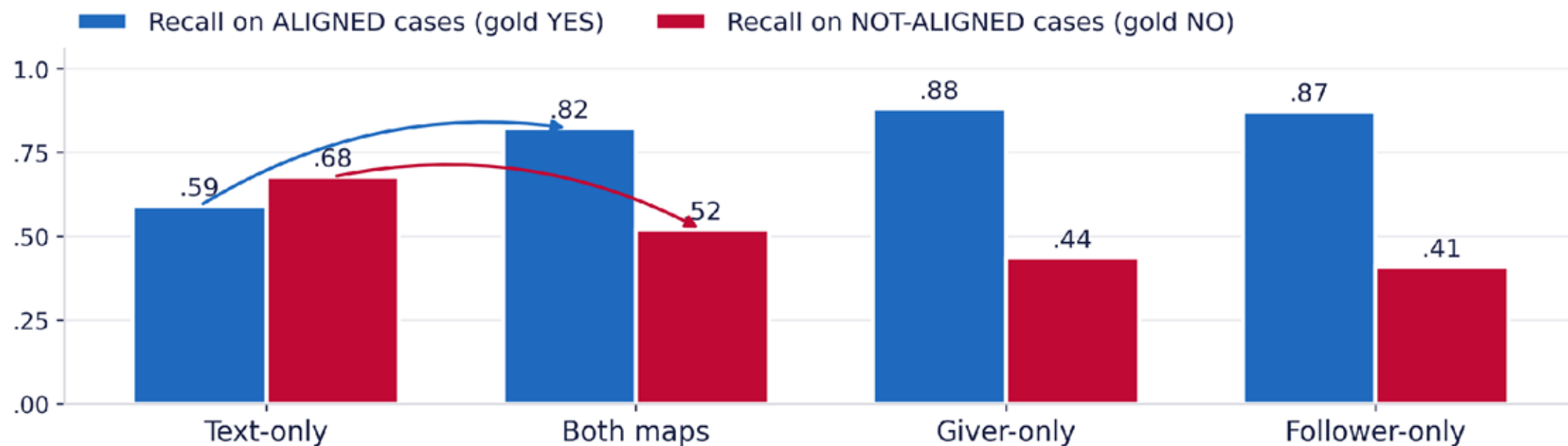
.590 → **.822**

RECALL · NOT ALIGNED

.677 → **.518**

YES-RATE

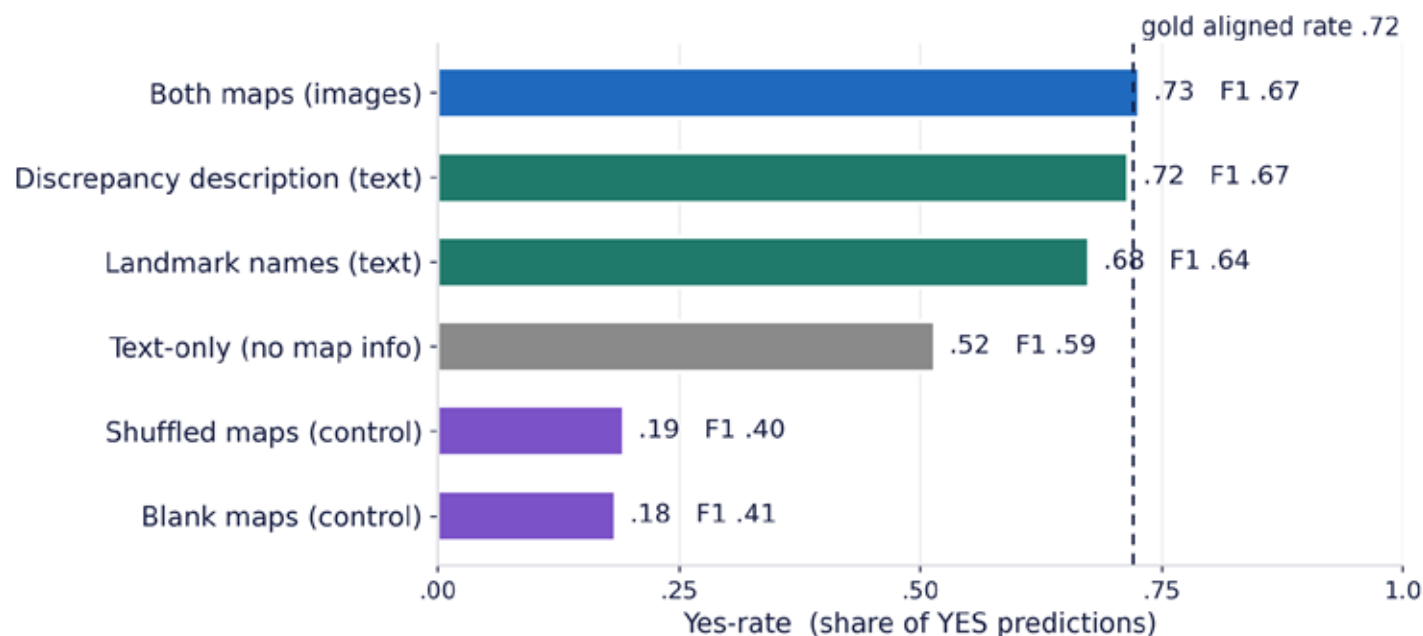
.515 → **.727**



Qwen3-VL-8B at startT. The gold aligned rate is .721 — the yes-rate alone is not the test; the direction of the recall trade-off is. Either single map triggers it more strongly; both maps moderate it, via cross-map discrepancy evidence.

FINDING 2

The bias travels with content, not with the channel



SAME CONTENT, NO IMAGE

Landmark names .675 and discrepancy text .716, against .727 for real maps.

IMAGES WITHOUT CONTENT

Blank .184, shuffled .193 — below even text-only (.515). The model turns hyper-conservative.

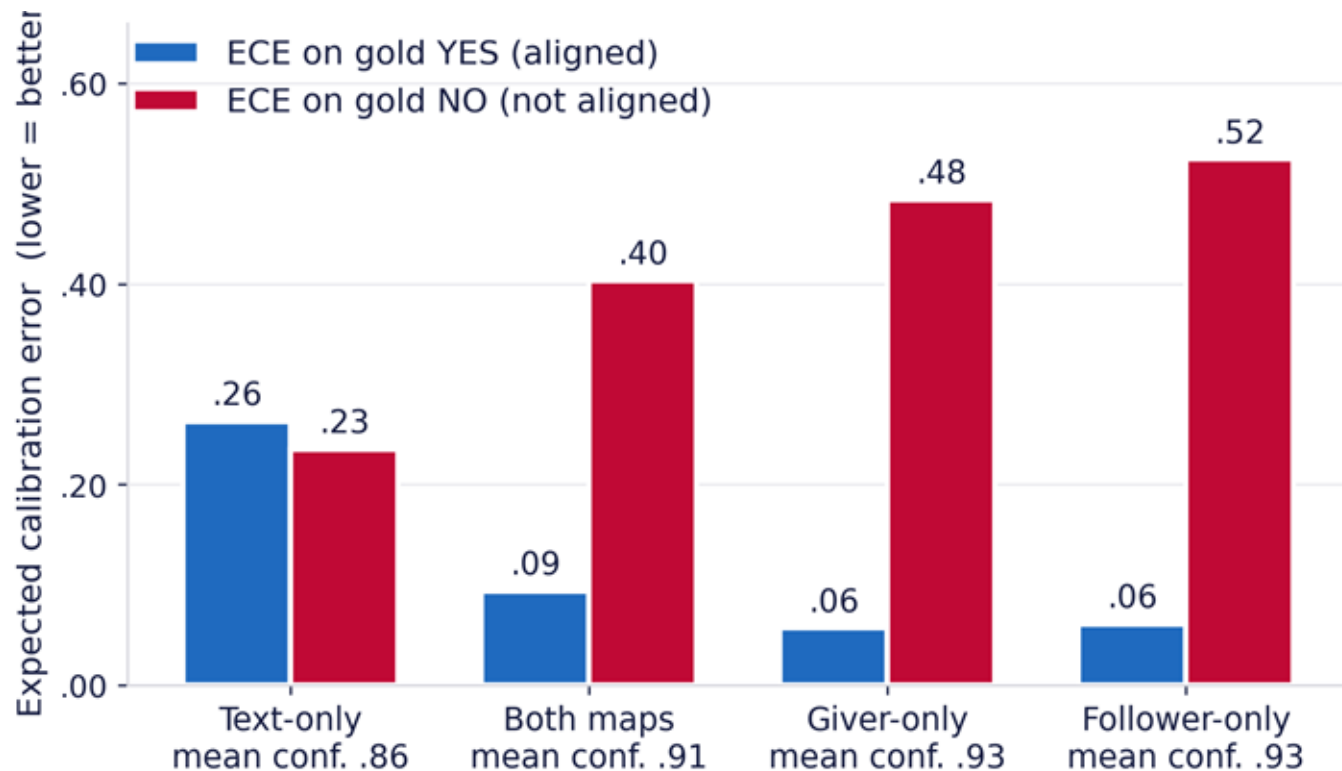
SO THE CUE IS MAP CONTENT

in pixels or in words.

All conditions: Qwen3-VL-8B at startT. Caveat: the textual conditions do not reconstruct the full spatial layout, and authentic images add a small further shift — so this bounds, but does not eliminate, a role for vision.

MECHANISM

On non-aligned cases, map access produces confident errors



CONFIDENT AND RIGHT ON GOLD YES

ECE .263 → .094 with both maps.

CONFIDENT AND WRONG ON GOLD NO

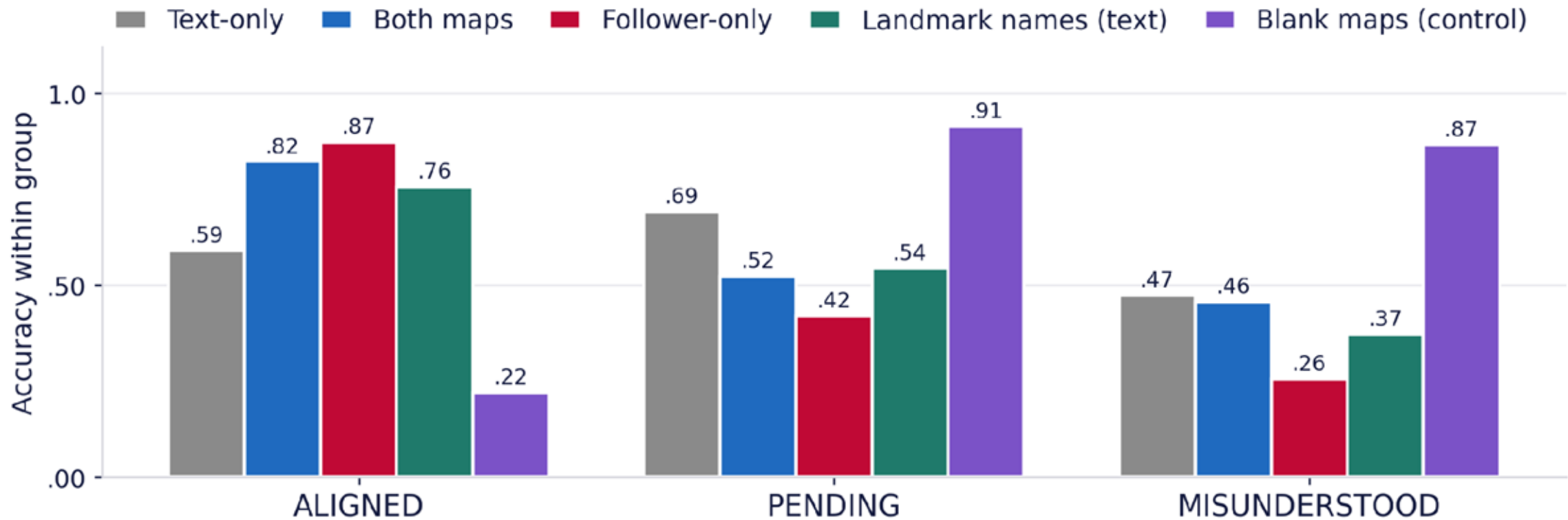
ECE .235 → .403, and .524 with the follower's map alone.

MORE THAN A THRESHOLD SHIFT

The gap between classes widens from .028 to .463.

Constrained single-token decoding, so confidence = $\exp(\log\text{prob})$. $n = 13,077$, Qwen3-VL-8B at startT. Lower ECE is better.

The aggregate gain is paid for by pending cases



n = 9,435

+23.2 pp .590 → .822

n = 3,403

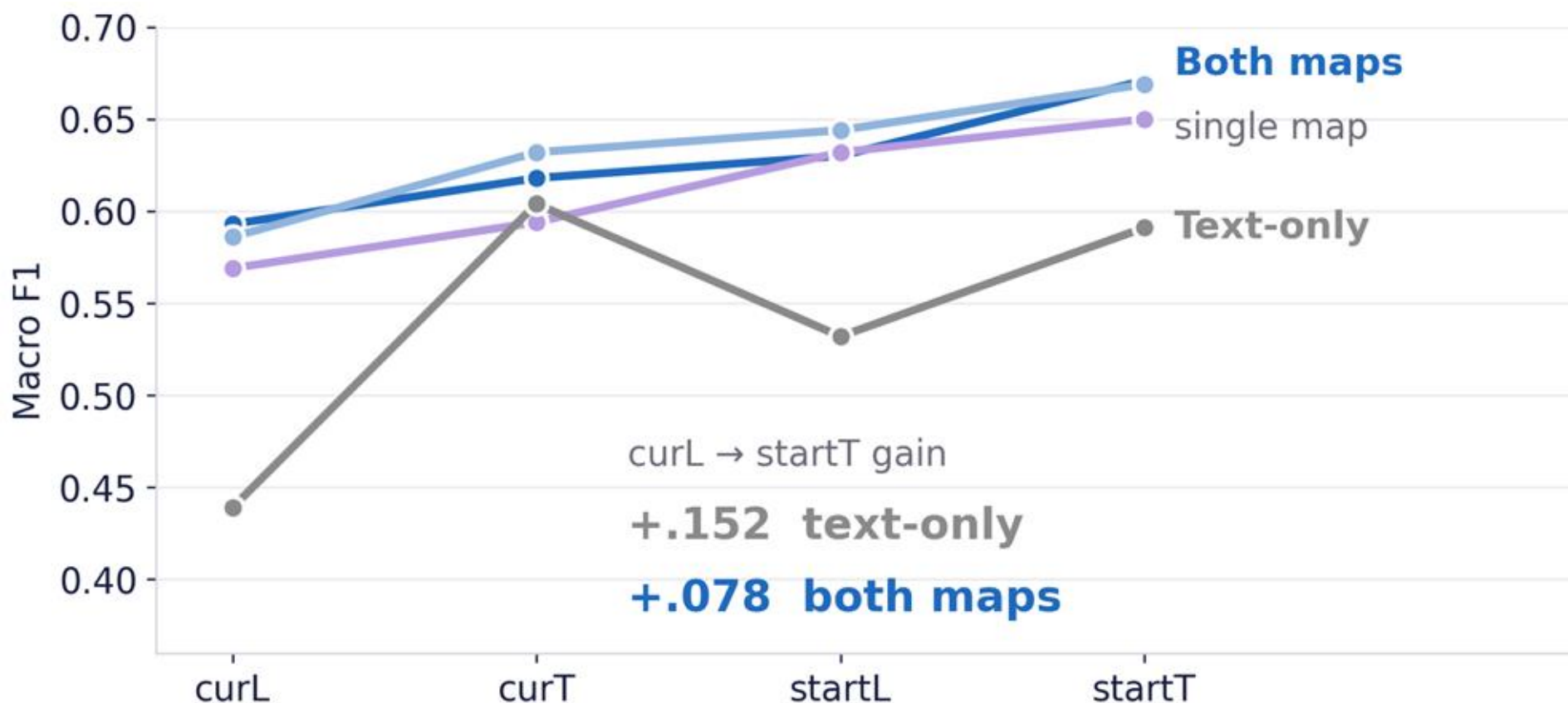
-16.8 pp $p < 10^{-6}$

n = 239

-1.7 pp n.s. $p = .724$

Text-only → both maps. Blank maps look strong on pending and misunderstood only because they answer NO to almost everything.

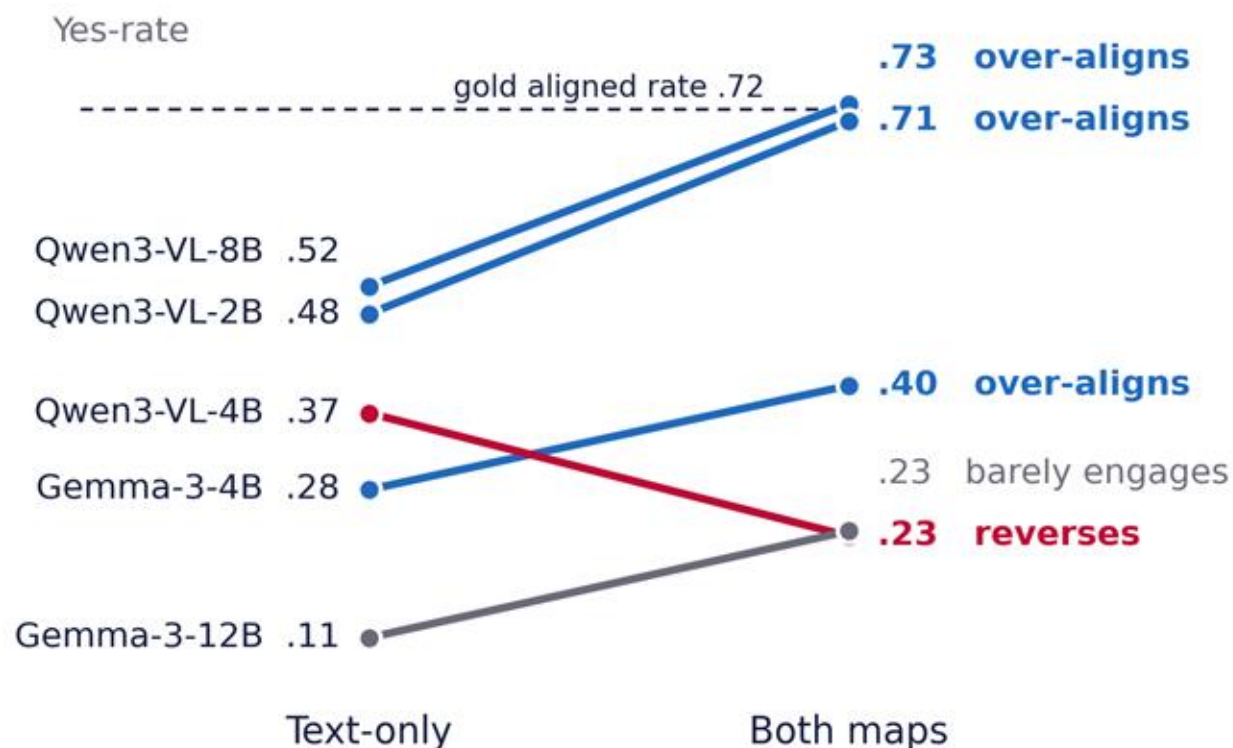
Dialogue evidence carries less weight once maps are in



Qwen3-VL-8B macro-F1 across the four context windows. Later turns always help — they carry repair and confirmation — but the gain halves once maps are available.

The reference-chain analysis is on backup slide B4, where the per-bucket gold aligned rate can be shown alongside it.

Model-dependent — and scale does not fix it



“SOME” IS DOING REAL WORK

Four of five shift upward, but only three land near the base rate. Gemma-3-12B stays extremely conservative; Qwen3-VL-4B reverses.

LARGER ≠ MORE DISCERNING

2B beats 4B (F1 .566 vs .411); Gemma-3-4B beats 12B (.510 vs .416).

ONE LIKELY CONTRIBUTOR

A model that cannot read the maps cannot show a content-driven bias.

Baseline grid at startT, n = 13,077. Gemma's SigLIP encoder pools every image to 256 tokens; in a map-reading check Gemma scored 81–82 F1 on landmark names against 88–90 for Qwen3-VL.

What the failure tells us about VLM grounding

SCENE EVIDENCE

what COULD be shared — landmarks
co-present on both maps

co-presence $\Rightarrow$ alignment



DIALOGUE EVIDENCE

what HAS been grounded —
recognition, confirmation, repair



under-weighted

THE MODEL'S JUDGEMENT

“ALIGNED” — because the landmark is on
both maps.

Potential common ground is read as established
common ground: the computational form of the
overhearer's illusion.

Evaluation should separate visual reference recognition from interactional common-ground tracking.

What this establishes — and what it does not

WHAT WE SHOW

- An interpretation-matching evaluation built on perspectivist annotations, over 13,077 reference expressions.
- Task-relevant map content shifts judgements toward YES — as pixels or as words.
- The cost concentrates on pending cases, and the errors there are confident.
- The pattern is model-dependent, and scale does not remove it.

WHAT WE DO NOT CLAIM

- Not an interactive test — the model can never ask or repair.
- The full condition grid is one model; the other four get the baseline comparison only.
- One corpus, MapTask-specific discrepancies, strong imbalance, and a rare misunderstood class ($n = 239$).
- Textual conditions omit full spatial layout; greedy binary decoding may mask softer uncertainty.

Next: interactive evaluation, further perspectivist corpora, and protocols that let a model express uncertainty and revise its judgement as the dialogue unfolds.

Seeing is not sharing

- 1 Maps improve macro-F1 by reading potential referential overlap as established common ground.
- 2 The bias travels with content, not channel — text reproduces it, content-free images reverse it.
- 3 The cost lands where grounding matters — confident errors on pending cases.

These models infer what the interlocutors could be talking about — not what they have established together.



Paper

aclanthology.org/2026.sigdial-1.49



Code + prompts

github.com/chnln/seeing-is-not-sharing



SINS dataset

huggingface.co/datasets/chnln/...

Baseline and modality results (Qwen3-VL-8B, startT)

Map information	F1-macro	R-pos	R-neg	Yes-rate
Text-only	.591	.590	.677	.515
Both maps	.671	.822	.518	.727
Giver-only	.669	.879	.436	.791
Follower-only	.650	.872	.408	.794
Landmark names (text)	.636	.756	.533	.675
Discrepancy description (text)	.668	.810	.528	.716
Blank maps	.407	.220	.910	.184
Shuffled maps	.402	.222	.884	.193

READING THE TABLE

R-pos = recall on aligned; R-neg
= recall on not-aligned.
Gold aligned rate: .721.
n = 13,077 for every row.

Tables 1 and 2 of the paper. Single-map conditions amplify the bias; both maps moderate it slightly via cross-map discrepancy evidence.

Accuracy by gold grounding status, with significance

Condition	Aligned	Pending	Misund.
Text-only	.590	.691	.473
Both maps	.822	.523	.456
Giver-only	.879	.441	.372
Follower-only	.872	.419	.255
Landmark names (text)	.756	.544	.372
Discrepancy desc. (text)	.810	.540	.368
Blank maps	.220	.913	.866
Shuffled maps	.222	.886	.858

MCNEMAR TESTS

Pending: $p < 10^{-6}$ for every authentic-map condition.

Misunderstood — both maps: $p = .724$ (n.s.).

Misunderstood — giver-only: $p = 8 \times 10^{-3}$; follower-only: $p = 3 \times 10^{-9}$.

Table 4. $n = 9,435 / 3,403 / 239$. Blank maps score high on pending and misunderstood only because they answer NO to almost everything (yes-rate .184).

Calibration by gold label

Condition	ECE	ECE-yes	ECE-no	Mean conf.
Text-only	.249	.263	.235	.863
Both maps	.174	.094	.403	.912
Giver-only	.172	.057	.484	.927
Follower-only	.185	.061	.524	.929

WHY THIS MATTERS

The class gap widens from .028 to .463 — spurious certainty on top of the threshold shift, not instead of it.

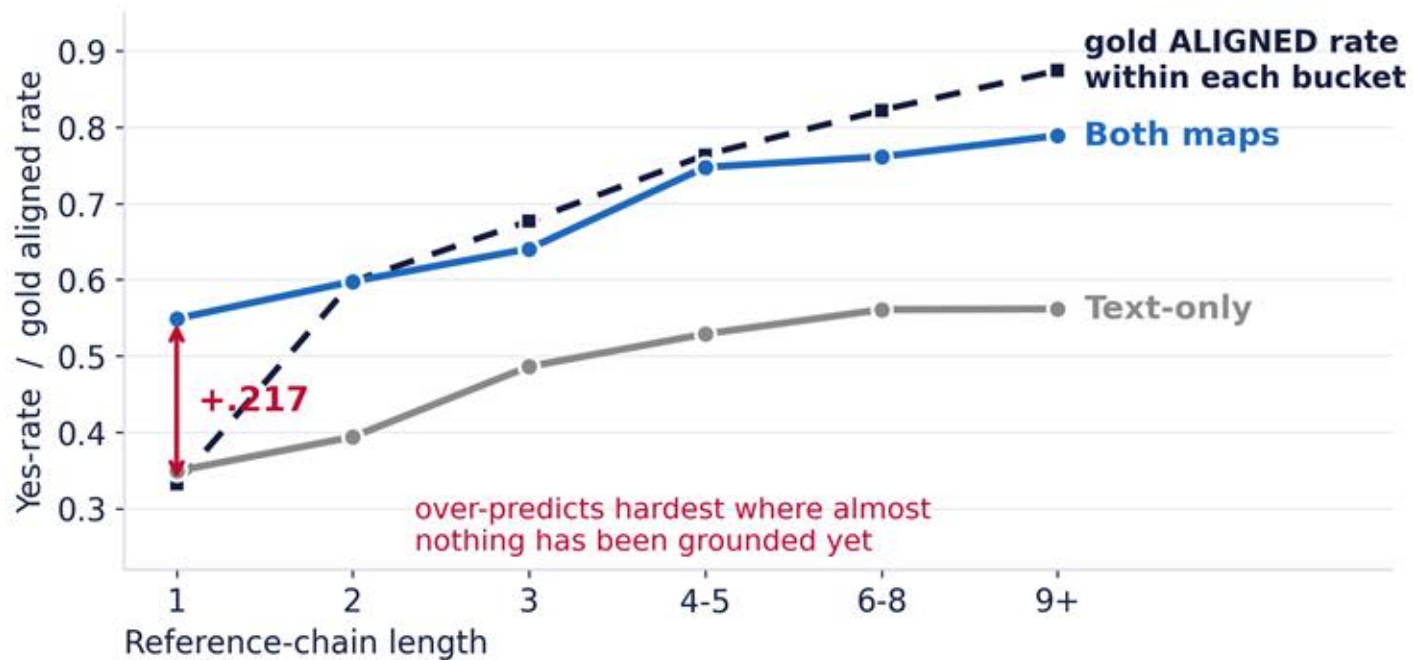
CROSS-MODEL CALIBRATION

Qwen3-VL-2B has the best ECE (.144) because it is the least confident (mean .789); Gemma-3-12B the worst (.531) at the second-highest confidence (.948).

Confidence rises with size in Qwen (.789 / .907 / .912) but falls in Gemma (.973 → .948). What holds in both families is that larger is not more discerning.

Table 3, n = 13,077, Qwen3-VL-8B at startT. $\text{conf} := \exp(\text{logprob})$ of the single generated YES/NO token.

Reference chains, against the right baseline



WHY NOT COMPARE TO .721

The chain subset is not the whole dataset: all 9,435 aligned REs sit in chains, but only 1,799 of 3,403 pending ones do.

And prevalence climbs within it, .33 → .87.

WHAT THE DATA SAYS

At one mention the model answers YES .55 of the time where only .33 are aligned — a 22-point over-prediction.

By 9+ mentions it sits slightly below the local rate.

1,665 chains covering 11,473 of the 13,077 REs. Suggestive, not causal: chain length is not randomly assigned. The full three-panel figure from the paper is Figure 2.

Why the models diverge

VISION ENCODING

Qwen3-VL: native dynamic-resolution encoder, hundreds of tokens for a 791×1024 map, 2D-RoPE spatial encoding.

Gemma-3: SigLIP, resized to 896×896 and average-pooled to 256 tokens per image regardless of complexity.

MAP-READING SANITY CHECK

All five models were asked to list landmark names and positions on all 32 maps.

Qwen3-VL 88–90 F1 · Gemma-3 81–82 F1, with character-level naming errors and spatial mislocations.

WHAT THIS DOES AND DOES NOT EXPLAIN

A model that cannot reliably extract map content cannot show a content-driven bias — so weak map reading plausibly explains Gemma's conservatism.

It does not explain Qwen3-VL-4B, which reads maps well and still reverses. We report the heterogeneity; a mechanistic account is future work.

Appendix E. The over-alignment bias is not a universal property of vision-language architectures.

Design choices and positioning

WHY ZERO-SHOT, GREEDY, BINARY?

To characterise default judgement behaviour, and because constrained single-token decoding makes the token probability a clean confidence signal.

Few-shot, CoT and thinking models are natural follow-ups.

IS THE TEXT/VISION COMPARISON FAIR?

The textual conditions convey landmark lists and discrepancies, not the full spatial layout — acknowledged in Limitations.

The direction of the argument is unaffected: text already reproduces the bulk of the bias.

RELATION TO WANG ET AL. (2025)

They measure the overall overhearer gap on referential dialogue.

We use perspectivist annotations to split that gap into potential vs. established common ground, and use modality controls to locate its source.

HUMAN BASELINE?

Gold labels come from the perspectivist annotation, not from a human-overhearer baseline.

A matched human-overhearer baseline is future work.

The same trade-off, grouped by condition

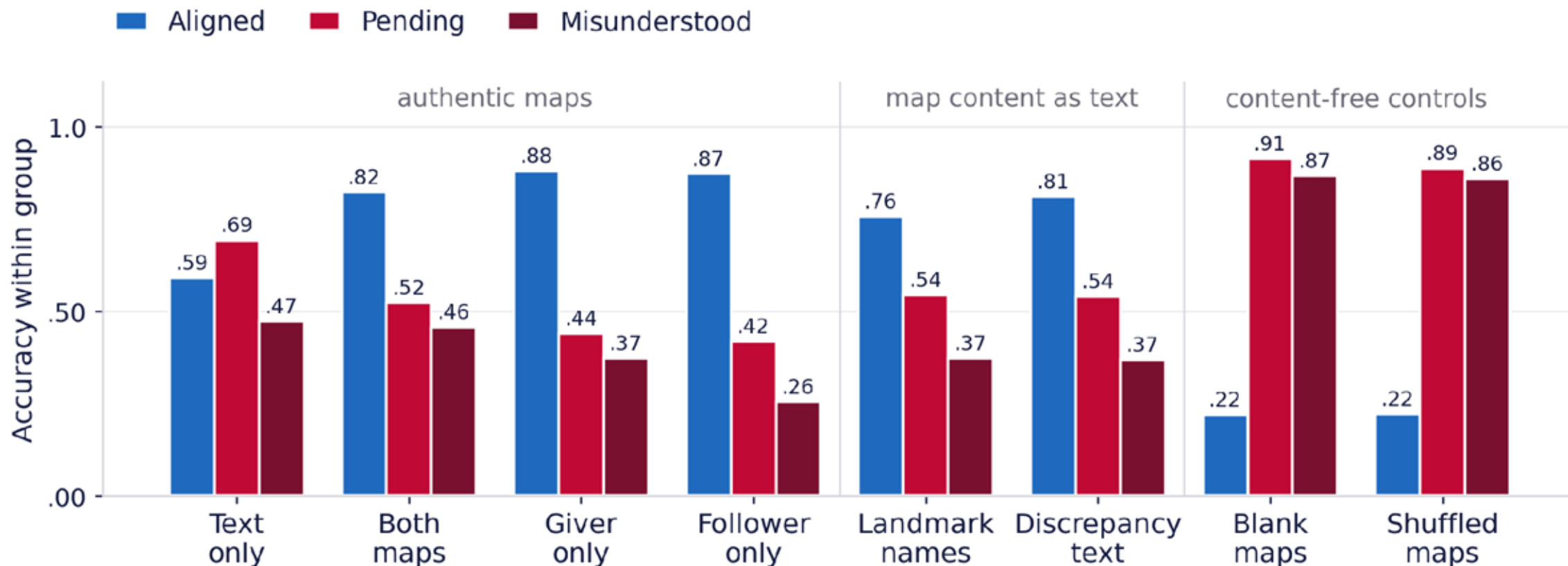


Table 4, all eight conditions. Anything carrying task-relevant map content — images or text — raises aligned accuracy and lowers pending and misunderstood accuracy.

The content-free controls invert the profile, for the wrong reason: they answer NO to almost everything.